

MANAGEMENT REVIEW

Management without theatre.

ROLE

DECISIONS UNDER PRESSURE

A floor that was too quiet, 156 decisions on the fireground, recognise then simulate, what stress does to choices, when a gut feeling deserves trust, and a card for a fast decision.



156

decision points of experienced fireground commanders: in fewer than 12% of them did they compare two or more options.

KLEIN, CALDERWOOD & CLINTON-CIROCCO, 2010

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Under pressure there is rarely time to list the options, weigh them and pick the best. Yet decisions still get made, and many of them are good. This issue asks how experienced people decide when time is short, what stress changes, and when a fast judgement can be trusted.

A fire lieutenant ordered his crew out moments before the floor collapsed, and at first could explain it only as ESP. In a study from 1985, experienced fireground commanders compared options in fewer than 12% of 156 decisions. Gary Klein's recognition-primed decision model shows what they did instead. Experiments show that stress makes choices riskier and the search for options shorter. Daniel Kahneman and Klein set out when such intuition deserves trust.

Stiven Janaqi EDITOR · STIVEN CATALYST

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HOW TO READ THIS ISSUE

Figure

Every figure has its source and year at the foot of its page.

Our reading

Where the editors interpret rather than the research, it says so.

Practice

The steps and the card are proposals to try, not research results.

COVER STORY

Something is wrong

A Cleveland fire commander told Gary Klein about a decision from his days as a lieutenant. The fire seemed to be in the kitchen of a one-storey house. Water hardly helped, and the crew pulled back into the living room. There he ordered everyone out. Moments later, the floor they had stood on collapsed.

The fire had been in the basement. Asked why he gave the order, he spoke of ESP. Klein went through the incident with him for two hours, and the reasons came out.

01**Water**

A kitchen fire should die down under water. This one did not.

02**Heat**

The room was hotter than a kitchen fire would make it.

03**Quiet**

For that much heat, the fire was too quiet. The floor muffled it.

OUR READING

He did not see more than his crew. He noticed sooner that what he saw did not match what he expected.

Source: Malcolm Gladwell, Little, Brown, 2005 (via Quoted in Fire Engineering, "Rapid Cognition: Think Fast!", 2011; Max Bazerman, Harvard Program on Negotiation, 2013)

From Klein's interviews, told after Malcolm Gladwell's retelling in Blink (2005) as retold by Fire Engineering and Max Bazerman; we could not see Klein's own account. The cues were reconstructed afterwards.

THE NUMBERS

One option, not two

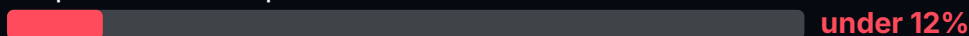
In 1985 Gary Klein, Roberta Calderwood and Anne Clinton-Cirocco interviewed 26 experienced fireground commanders (23 years of experience on average) about 156 decision points in real incidents. They expected them to compare at least two options. They were wrong.

156 DECISION POINTS OF FIREGROUND COMMANDERS, STUDY OF 1985

Recognised a typical situation and its usual action



Compared two or more options



Usually one option came to mind first, and it was enough. In 2021 Klein added that inexperienced decision makers use this approach less than half the time, and that people use it even when time is not short.

OUR READING

Under pressure, experts do not choose faster among options. Mostly they never line options up at all.

Sources: Gary Klein, Roberta Calderwood & Anne Clinton-Cirocco, Journal of Cognitive Engineering and Decision Making 4(3), 2010 (via Abstract at SAGE); Gary Klein, Psychology Today, 2021

Done in 1985, republished with a postscript in 2010; the abstract gives the shares only as "over 80%" and "less than 12%", and so does the chart. Interviews after the event: how commanders remembered deciding.

THE MODEL

Recognise, then simulate

From these interviews came the recognition-primed decision (RPD) model. Klein describes two processes: matching the situation to ones met before, which points to a workable action, then imagining how it would play out.

THE RPD STRATEGY, AFTER KAHNEMAN & KLEIN (2009)

01 Recognise The situation matches a pattern; a plausible option comes first.	02 Simulate Run the option forward in the mind: will it work here?	03 Act or adjust If it works, act. If it has a flaw, change it.	04 Next option If it cannot be fixed, take the next most plausible one and run it the same way.
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With patterns from more than a decade of fires, the commanders could foresee how flames would spread and notice signs that a house was about to collapse. Klein calls recognition fast and intuitive, simulation slower and deliberate.

OUR READING

The check is built in: the mental simulation. A fast decision is not thoughtless; the thinking goes into a single option.

Sources: Daniel Kahneman & Gary Klein, *American Psychologist* 64(6), 2009; Gary Klein, *Psychology Today*, 2021

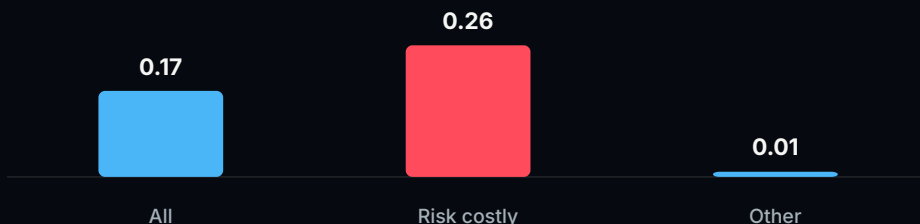
Steps after Kahneman and Klein's description of the commanders, worded by the editors. The model describes experienced people; it does not promise that the first option is right.

WHAT THE RESEARCH SAYS

What stress does to choices

Katrin Starcke and Matthias Brand pooled laboratory studies that induced stress before a decision: 32 datasets, 1,829 participants. Overall, decisions under stress were more disadvantageous, more reward-seeking and riskier than without it.

EFFECT OF STRESS ON DECISIONS (D), 2016



The effect was clear where risk did harm and absent elsewhere. In an older experiment (1987), Giora Keinan gave 101 students decision problems with no time limit. Under stress, they more often offered a solution before considering all the alternatives, and scanned them less systematically; how they scanned was linked to getting the answers right.

OUR READING

Stress does not only hurry a decision. Even without a deadline, it makes people stop looking sooner. With a trained eye that may be enough; without one, the first idea goes untested.

Sources: Katrin Starcke & Matthias Brand, *Psychological Bulletin* 142(9), 2016 (via Abstract at PubMed); Giora Keinan, *Journal of Personality and Social Psychology* 52(3), 1987 (via Abstract at PubMed)

d is an effect size; by the usual convention 0.2 is small. Laboratory studies with stress induced on purpose, not crews in the field. Keinan's study is from 1987.

HOW IT IS MEASURED

When the gut deserves trust

Daniel Kahneman spent much of his career finding intuition flawed in experiments; Klein, making the case for expert intuition. In 2009 they set out together when it can be trusted. Their answer looks at the situation, not at how sure one feels:

01

A regular environment

Stable links between visible cues and what follows: firefighting, medicine. Not the price of a single stock.

02

A chance to learn it

Prolonged practice and feedback that is rapid and unequivocal.

Confidence, they write, is no reliable sign that an intuition is valid, and expertise is fractionated: skilled in some tasks of a job, not in others. In 2010 Klein added: take the gut feeling as a data point, then check whether it makes sense here.

HYPOTHETICAL EXAMPLE, A SHIFT LEAD'S MONTH

Held 16 of 22 fast decisions went as expected

Missed 6, of which 5 in situations rarely met before

The count shows where experience is real and where it only feels real. The numbers are invented.

Sources: Daniel Kahneman & Gary Klein, *American Psychologist* 64(6), 2009; Daniel Kahneman & Gary Klein, interviewed by Olivier Sibony & Dan Lovallo, *McKinsey Quarterly*, 2010

Conditions and examples from Kahneman & Klein (2009), Klein's words from *McKinsey Quarterly* (2010). Writing the expectation down first is the editors' way to get rapid feedback.

[More in the essay](#)

When to solve it yourself and when to take it higher

TOOL OF THE ISSUE

The fast decision card

One card, one decision. The first four lines before you act or right after, the last two when the pressure is off. An expectation that failed is the most useful line.

01 What I see cues and facts, not interpretations

02 What I expect what should happen next if I read it right

03 What I will do the first workable option, run through once in my head

04 What would tell me I am wrong the sign that makes me stop and look again

05 What happened as expected? If not, what did I miss

06 Handed over what stays open, who owns it, who was told

Sources: Gary Klein, *Psychology Today*, 2021; Daniel Kahneman & Gary Klein, *American Psychologist* 64(6), 2009; Daniel Kahneman & Gary Klein, interviewed by Olivier Sibony & Dan Lovallo, *McKinsey Quarterly*, 2010

A practice proposed by the editors, after the RPD model, Kahneman and Klein's conditions and Klein's advice to treat a gut feeling as a data point. The handover tool keeps open items with an owner.

Open the tool: [Shift Handover](#)

stivencatalyst.com/tools/shift-handover/

SOURCES AND METHOD

Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.

Blink: The Power of Thinking Without Thinking

Malcolm Gladwell, Little, Brown, 2005 · via Quoted in Fire Engineering, "Rapid Cognition: Think Fast!", 2011; Max Bazerman, Harvard Program on Negotiation, 2013

Rapid Decision Making on the Fire Ground: The Original Study Plus a Postscript

Gary Klein, Roberta Calderwood & Anne Clinton-Cirocco, Journal of Cognitive Engineering and Decision Making 4(3), 2010 · via Abstract at SAGE
<https://doi.org/10.1518/155534310X12844000801203>

The RPD Model: Criticisms and Confusions

Gary Klein, Psychology Today, 2021
<https://www.psychologytoday.com/us/blog/seeing-what-others-dont/202102/the-rpd-model-criticisms-and-confusions>

Conditions for intuitive expertise: A failure to disagree

Daniel Kahneman & Gary Klein, American Psychologist 64(6), 2009
<https://doi.org/10.1037/a0016755>

Effects of stress on decisions under uncertainty: A meta-analysis

Katrin Starcke & Matthias Brand, Psychological Bulletin 142(9), 2016 · via Abstract at PubMed
<https://doi.org/10.1037/bul0000060>

Decision making under stress: Scanning of alternatives under controllable and uncontrollable threats

Giora Keinan, Journal of Personality and Social Psychology 52(3), 1987 · via Abstract at PubMed
<https://doi.org/10.1037/0022-3514.52.3.639>

Strategic decisions: When can you trust your gut?

Daniel Kahneman & Gary Klein, interviewed by Olivier Sibony & Dan Lovallo, McKinsey Quarterly, 2010
<https://www.mckinsey.com/capabilities/strategy-and-corporate-growth/our-insights/strategic-decisions-when-can-you-trust-your-gut>

EDITORIAL METHOD

Each figure was checked for its year, its publisher and what exactly it measures. Where the publisher's page could not be opened, the figure was checked against independent summaries and is marked "via". The editors' interpretation is marked "Our reading". Figures that could not be confirmed are not in the issue.



MANAGEMENT REVIEW

Management without theatre.

Every issue, one management question, checked against the best research.

ALL ISSUES

stivencatalyst.com/magazine/management-review.html